Internal Assessment 01

Full Stack Web Development - Question Paper

Name: _____ Roll No: _____ Date: _____

Max Marks: 30 | Duration: 90 minutes | Units: Week 1 to Week 3 (HTML/CSS, JavaScript, Node/Express)

Instructions

- Answer all questions. Build everything inside src/.
- Save a screenshot of the working output in screenshots/ as proof.
- Use only Node.js, Express and plain HTML, CSS and JavaScript.

Problem: Split Bill Calculator

A group of friends wants to split a restaurant bill equally, including an optional tip.

Q1. Page design (10 marks)

Create a responsive page with a form containing: bill total, number of people and tip percentage, and a Calculate button. Use semantic HTML, a Flexbox or Grid layout and a separate stylesheet.

Q2. Express API (10 marks)

Create POST /api/split that accepts total, people and tipPercent and returns the tip, grand total and amount per person rounded to two decimals. Return 400 with a clear message when total is not positive, people is not a whole number of at least 1, or tipPercent is outside 0-100. Serve the page from the same Express app.

Q3. Integration (10 marks)

Use fetch with async/await to call the API when the form is submitted. Show the result in a table, show API errors on the page without reloading, and hide the result when an error occurs.

Deliverables

- src/ (server.js, package.json, public/index.html, style.css, script.js)
- screenshots/exam-output-proof.png